

An Explainable Machine Learning Framework for Energy Forecasting, Anomaly Detection, and Cost-Aware Decision Support in Smart Campuses

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Abstract—A consumption forecast does not tell a campus facilities office what to switch off, what the waste costs, or which repair to fund first. Building-energy research has developed each analytical stage separately; integration into an operational decision remains less studied. This paper presents CampusWatt, a framework in which forecasting is treated as a reference rather than a product: the forecast establishes what consumption the operating context explains, the residual identifies what it does not, an audited equipment inventory converts that residual into a shortlist of physically admissible causes, and a high-tension tariff model prices the result so competing interventions can be ranked in currency. Readings arrive through three paths — bulk upload, daily entry and hourly entry — reconciled under a deterministic policy that records provenance and raises disagreement as an explicit conflict. Evaluation uses a synthesised year of hourly data for a five-zone academic block, partitioned chronologically with the final quarter held out. Zone-level gradient boosting achieves 5.6 per cent MAPE in the computer laboratories against 72.0 per cent for a recomputed seasonal-naïve baseline, and improves on that baseline in every zone. Detection was evaluated against eleven labelled fault events using a reproducible reference predictor: nine of eleven events were surfaced, while hour-level precision remained low across all thresholds examined, which is reported explicitly because the detection experiment characterises the reference predictor rather than the trained models. Tariff modelling attributes 50.4 per cent of modelled annual cost to the maximum-demand component. The dataset is synthesised from an audited inventory and no metered validation was performed; the evidence supports framework feasibility, not verified savings.

Keywords—building energy forecasting, explainable artificial intelligence, anomaly detection, gradient boosting, maximum demand charge, decision support, smart campus

I. INTRODUCTION

Electricity is among the largest recurring operating costs an Indian educational institution carries, and among the least instrumented. A campus of several buildings is commonly served by one high-tension connection and billed monthly against a single meter reading. That reading is a total. It records how much was consumed and says nothing about

which part of the building consumed it, during which hours, or for what reason.

The delay compounds the problem. By the time a bill arrives, the behaviour that produced it has been repeated for four to six weeks. Air-conditioning left running through a vacation, a compressor losing efficiency, a laboratory whose machines never sleep — none announce themselves. Each is absorbed into a monthly figure that looks unremarkable beside the previous month.

Building-energy forecasting has been studied extensively, with statistical and machine-learning approaches surveyed across several reviews [1], [3], [4]. Operational decision support after the forecast is less integrated. A facility manager told that Tuesday night was anomalous still does not know which equipment drew the power, what the incident cost, or whether fixing it is worth more than the three other repairs competing for the same maintenance budget.

A. Problem Statement

Campus energy is managed without the information required to manage it. Consumption is measured at a granularity too coarse to locate waste, reported on a cycle too slow to correct it, and priced by a model too simple to rank the available responses. A system resolving only the first of these produces dashboards that are not acted upon.

A second constraint shapes what can realistically be deployed. Sub-metering is usually absent, and the data that does exist arrives irregularly: an occasional export from a utility portal alongside readings a staff member writes in a register. A design assuming a continuous instrumented feed cannot be adopted where none exists; a design accepting only typed readings cannot exploit whatever history is available. Supporting both, and reconciling them when they disagree, is a requirement rather than a convenience.

B. Research Objectives

- O1 — Forecast zone-level consumption accurately enough to serve as a detection reference, using weather and academic-calendar inputs only.
- O2 — Attribute a deviation from that forecast to specific installed equipment using an audited inventory, without sub-metering.
- O3 — Determine whether modelling the complete high-tension tariff, rather than a flat energy rate, changes the intervention ranking.
- O4 — Reconcile readings arriving through independent paths at different granularities, preserving provenance and surfacing disagreement.
- O5 — Quantify forecasting performance against a reproducible seasonal-naïve baseline, and assess whether the pipeline surfaces labelled injected fault events.

C. Significance

Institutional energy reduction is usually framed as a capital problem, and those measures compete for budgets that rarely exist. This work investigates whether analysis can identify operational opportunities without capital expenditure, by making consumption visible at a granularity where its causes become identifiable. The constraint addressed — a large building, one meter, no analytics budget — is representative of institutions where fine-grained sub-metering has not been installed, and the work aligns with Sustainable Development Goal 12.

D. Contributions

The contribution is system-level integration and operationalisation rather than a new learning algorithm. Four elements are claimed.

- A reconciliation policy for heterogeneous energy inputs assigning each zone-day one of five provenance states and escalating disagreement beyond a defined tolerance to a human decision.
- An asymmetric validation rule rejecting physically impossible typed values while admitting the same values from imported files, since the second may be the fault the system exists to detect.
- An inventory-bound attribution procedure converting a residual in kilowatts into an enumerated set of physically admissible equipment hypotheses, eliminating those requiring more units than are installed.
- An evaluation using labelled synthetic faults, permitting event-level detection to be measured against ground truth in a setting where metered anomaly labels do not exist.

II. LITERATURE REVIEW AND RESEARCH GAP

A. Forecasting, Detection and Explainability

Statistical and learning-based load forecasting has been surveyed across several reviews [1]. Gao et al. applied metaheuristic optimisation to an extreme learning machine and reported gains over conventional variants [2]; Liu et al. compared nine model families across fifty-two building descriptors and found feature composition governs accuracy more than model choice [3], which informed the feature design in Section III-E; Qiao et al. addressed sparse instrumentation using Boruta selection with empirical mode decomposition [4]. Campus-specific work is thinner, covering occupancy-and-weather prediction for a university building [5] and multi-year meter collections [6]. This work adopts the established approach without extending it, and re-uses the forecast as a detection reference rather than as the deliverable.

For detection, Himeur et al. identified prediction-residual methods as a dominant paradigm [7] and Araya et al. proposed an ensemble framework [8]. Telila et al. thresholded prediction error over smart-meter data [9]; their use of deliberately injected faults to obtain labels is the practice adopted here. These approaches produce a flagged interval, and this work adds the stage converting that interval into an equipment hypothesis. On explainability, Noorchenarboo and Grolinger conditioned SHAP background sets on each anomaly [10], Moon et al. applied Shapley attribution to educational buildings [11], and Chung and Liu compared LIME against SHAP [12]. All explain a model in terms of its inputs; a maintenance technician needs an explanation in terms of equipment, and this work implements both as separate artefacts.

B. Demand-Side Management

Balasubramanian et al. applied peak clipping and load shifting at an Indian university campus and showed billing cost falling without service reduction [13]. Patil and Tanavade documented staged interventions including lighting replacement and timetable-mapped switching [14], and Li et al. examined semester scheduling as an indirect demand-response lever [15]. These treat scheduling as a manual planning exercise; this work computes the recommendation from a forecast and prices it against alternatives.

C. Research Gap

No individual stage is unstudied, and this work claims no algorithmic novelty in any of them. The gap addressed is that the stages are seldom joined, and the joins are where analytical output becomes an institutional decision: forecast to residual, residual to equipment, equipment to cost, cost to ranked action. Two further omissions recur in the reviewed work. Many energy-analytics studies simplify monetary impact to an energy-rate calculation, which can omit the demand-charge component and with it the class of interventions that reduce billed peak without reducing consumption. And data acquisition is commonly assumed to

be a single clean instrumented feed, which excludes the institutions where the problem is most acute.

III. METHODOLOGY

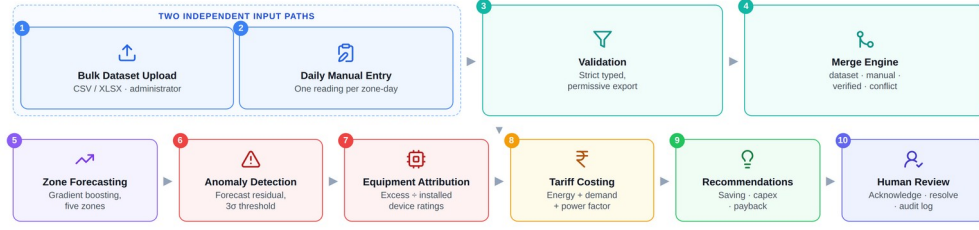


Fig. 1. The processing pipeline: multi-path acquisition, reconciliation, forecasting, detection, attribution, pricing and human review.

A. Study Site and Zone Partition

The study unit is one academic block of a college in Chennai, partitioned into five functional zones rather than treated as a single meter. Partitioning follows usage pattern rather than physical adjacency, because rooms sharing a usage pattern share a consumption profile and are jointly predictable. A physical audit established the connected load of each zone, given in Table I, which bounds what any reading for that zone can legitimately be.

TABLE I. ZONE PARTITION AND AUDITED EQUIPMENT INVENTORY.

Zone	Rooms	Connected load	Share
Computer Labs	2	39.6 kW	48.7%
Classrooms	23	22.8 kW	29.1%
Server Room	1	6.2 kW	10.0%
Common Areas	5	2.8 kW	6.4%
HOD Rooms	3	6.9 kW	5.7%

The distribution is uneven to a degree that governs the analysis: two laboratories account for almost half of block consumption.

B. Dataset Construction

Sub-metering was not available and the institutional meter reports monthly aggregates only. Rather than substitute a public dataset from an unrelated building, the series was synthesised from the block's own equipment census, a Chennai climate model and the institutional academic calendar. Hourly consumption is computed as installed device power multiplied by a duty cycle conditioned on hour, day type and weather:

$$E(z,t) = \sum_i P_i \cdot n_i(z) \cdot d_i(t, \theta_t, c_t) / 1000 \quad (1)$$

where d_i is the duty cycle of device type i at hour t given temperature θ_t and day-type c_t . Duty cycles are physically motivated rather than fitted: fan operation rises with temperature, lighting follows occupancy modified by daylight, laboratory machines follow scheduled sessions with a standby draw outside them, and server load is

The framework is organised as a pipeline in which each stage consumes the output of the previous one and adds a distinct interpretation. Fig. 1 shows the arrangement.

constant. Table II lists the generator inputs so the series can be regenerated.

TABLE II. GENERATOR INPUTS FOR THE SYNTHESISED SERIES.

Generator input	Specification
Equipment inventory	5 zones, per-device rated power and counts
Weather model	Chennai seasonal and diurnal temperature, humidity
Academic calendar	2 semesters, 2 exam periods, 2 vacations, holidays, working Saturdays, event days
Duty-cycle rules	Per device type, conditioned on hour, day type, temperature
Anomaly categories	7 (overnight, weekend, vacation, holiday, baseline creep, equipment fault, meter dropout)
Anomaly injection	11 events, 532 affected hours, recorded in a withheld ground-truth file
Output	43,800 rows, 147,155 kWh, peak 162.7 kW, calendar year 2025

Because the faults are known, detection can be measured against ground truth. Public consumption datasets carry no anomaly labels, which is why detection performance in much of the reviewed literature is reported qualitatively.

C. Multi-Path Acquisition and Validation

Readings enter through three paths: bulk file upload by an administrator, one daily total typed by a block manager, and one hourly reading typed as the meter is read. The third exists because a daily total cannot say when consumption occurred, and equipment running between 02:00 and 05:00 disappears entirely into it. The physical ceiling for a zone follows from the inventory:

$$C_z = (\sum_i P_i \cdot n_i(z) \cdot 24 / 1000) \cdot \lambda \quad (2)$$

Here $\lambda = 1.25$ is a headroom multiplier allowing for meter tolerance and unrecorded equipment, so C_z is a deliberately permissive bound rather than a nameplate sum. Validation is deliberately asymmetric. A typed value above C_z is rejected, since a figure exceeding what the installed equipment can draw is almost certainly a keying error. The same value arriving in an imported file is accepted and flagged, because it may be a genuine fault — the condition the detector exists

to find. Rejecting it would suppress the evidence the system was built to surface.

D. Reconciliation Policy

TABLE III. RECONCILIATION POLICY APPLIED PER ZONE-DAY.

Condition	State	Value used
Dataset only	dataset	Imported series
Manual only	manual	Total, spread by profile
Both, within τ	verified	Manual total, dataset shape
Both, beyond τ	conflict	Manual, pending decision
Administrator ruled	resolved	Chosen source, audited

Where sources agree within tolerance the manual total is preferred for the daily quantity while the imported file supplies the hourly shape. Disagreement beyond tolerance is not averaged away: it is recorded as a conflict and an administrator must rule.

E. Forecasting

A separate histogram-based gradient boosting regressor is trained per zone, because the zones differ by more than an order of magnitude in scale and are driven by different mechanisms. Thirty-eight features were engineered across four families: autoregressive lags at 1, 2, 24, 48 and 168 hours; rolling means and deviations shifted to exclude the current observation; cyclical encodings of hour and weekday; and exogenous weather terms including cooling degrees above a 25 °C base. The five models are trained on January–September and evaluated on October–December, held out entirely.

The most consequential feature is a learned profile giving mean consumption for each hour and day-type combination, fitted on training observations only. When the calendar moves from ordinary teaching into an examination period, every lag feature is drawn from ordinary working days and misleads the model at precisely the moment prediction is hardest; the profile supplies a direct estimate independent of the recent past. Table IV summarises the configuration and evaluation setup.

TABLE IV. FORECASTING CONFIGURATION AND EVALUATION SETUP.

Parameter	Value
Estimator	HistGradientBoostingRegressor
Models / features	5 (one per zone) / 38
Max iterations / depth	700 / 7
Learning rate	0.05
Train / test split	Jan–Sep / Oct–Dec, chronological
Baseline	Seasonal-naive, lag 168 h, recomputed

F. Detection and Equipment Attribution

An observation is flagged when its standardised residual exceeds a threshold k of training standard deviations:

$$|r(z,t)| / \sigma_z > k, \quad r(z,t) = E(z,t) - \hat{E}(z,t) \quad (3)$$

Because the reference already accounts for weather, hour and day type, a residual represents consumption the operating context does not explain, so a working Monday is not reported as anomalous merely for differing from the preceding Sunday. Residual thresholding is insensitive to slow drift, since a gradually rising baseline is absorbed into the lag features; a rolling-baseline monitor therefore runs alongside it.

Attribution converts the residual into a physical hypothesis. For an excess ΔP in zone z , the units of device type i required to account for it are:

$$u_i = 1000 \cdot \Delta P / P_i, \quad \text{admissible iff } u_i \leq n_i(z) \quad (4)$$

Hypotheses demanding more units than are installed are eliminated. What survives is a shortlist a technician can check on site. A 19 kW overnight excess in the laboratories is numerically consistent with approximately 127 desktop-equivalent loads at 150 W, admissible against 240 installed, and is not consistent with ceiling fans, which would require 253 against 30 installed. Block-level metering cannot establish that desktops caused the excess; the output is a hypothesis set, not a diagnosis.

G. Cost Model

Cost is computed over the full high-tension tariff rather than a flat rate, using the indicative published constants in Table V.

TABLE V. TARIFF CONSTANTS. VALUES ARE INDICATIVE PUBLISHED FIGURES, NOT THE INSTITUTION'S CONTRACTED RATES.

Constant	Value	Basis
Energy rate	₹8.05 / kWh	Indicative HT commercial
Demand rate	₹580 / kVA / month	Indicative HT commercial
Sanctioned demand	150 kVA	Site connection
Power factor / target	0.94 / 0.95	Modelled / tariff target

$$S_{max} = P_{max} / \cos \phi \quad (5)$$

$$C_{annual} = E_y \cdot c_e + 12 \cdot S_{max} \cdot c_d \quad (6)$$

where E_y is annual consumption (kWh), c_e the energy rate (₹8.05/kWh), P_{max} the recorded peak real power (kW), $\cos \phi$ the modelled power factor (0.94), S_{max} the corresponding apparent power (kVA) on which the demand charge is levied, and c_d the demand rate (₹580/kVA/month). The demand charge is billed on apparent power, so the recorded peak of 162.7 kW corresponds to 173.1 kVA at the modelled power factor. The power-factor penalty and statutory duty are implemented in the platform but are excluded from (5) and from the composition reported in Section IV-D, which covers the two dominant components only.

H. Implementation and Evaluation Protocol

The framework is implemented as a service-oriented application with a Python service layer, a relational store

under versioned migrations and a browser dashboard. Access is role-differentiated and enforced server-side, every state-changing operation is audited, and the service layer carries 371 automated tests at 90 per cent statement coverage.

The evaluation protocol was fixed before results were examined. No shuffling is performed, since random partitioning of a time series allows information from the future to inform predictions of the past. Hours marked anomalous are excluded from training so residuals remain interpretable as deviation from normal operation. Accuracy is reported as MAPE alongside mean absolute error, because MAPE normalises by the actual value and exaggerates error in low-consumption zones. The seasonal-naive baseline was recomputed directly from the released series rather than quoted, so the comparison is reproducible.

IV. RESULTS

A. Load Characterisation

The working-day load profile has twin peaks at 12:00 and 15:00 separated by a lunch trough, corresponding to the two laboratory sessions. Vacation load is roughly a fifth of teaching load but does not fall to zero, and the block draws 4.5 kW at 03:00 with nobody present — a standing load delivering no service, equivalent to about 39,400 kWh across a year if sustained.

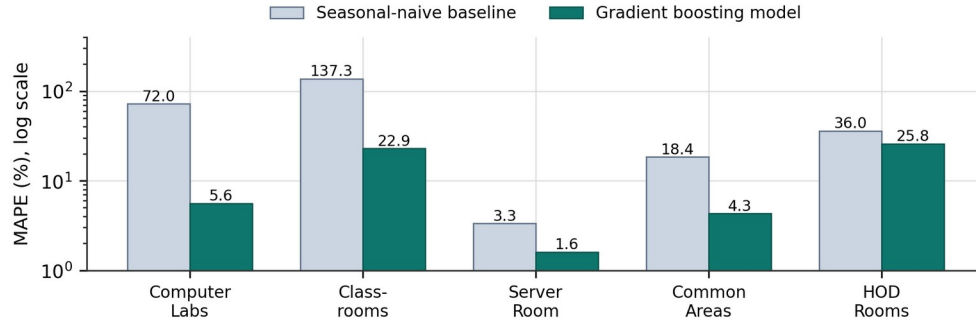


Fig. 2. Per-zone forecast MAPE against the recomputed seasonal-naive baseline, logarithmic scale, over non-anomalous test hours.

C. Detection Evaluation

Detection was evaluated against the eleven labelled events using a reference predictor that can be recomputed from the released series: a per-zone mean conditioned on day type and hour, fitted on non-anomalous training hours. This predictor, not the trained models, was used because the deployed service returns a reference profile rather than live model inference (Section V-C). The figures below therefore characterise the reference predictor and should not be read as an estimate of the detection performance of the trained models, which was not evaluated.

B. Forecasting Accuracy

TABLE VI. FORECAST ACCURACY BY ZONE. MAE IN kWh PER HOUR; NAIVE IS THE RECOMPUTED SEASONAL-NAIVE BASELINE MAPE; CUT = (NAIVE – MODEL) / NAIVE × 100.

Zone	MAPE	MAE	Naive	Cut
Computer Labs	5.6%	0.250	72.0%	92%
Classrooms	22.9%	0.733	137.3%	83%
Server Room	1.6%	0.023	3.3%	52%
Common Areas	4.3%	0.047	18.4%	77%
HOD Rooms	25.8%	0.102	36.0%	28%

Every zone improves on the baseline; Fig. 2 presents the comparison on a logarithmic scale. Block-level accuracy from the offline evaluation is 5.75 per cent MAPE with mean absolute error of 0.96 kWh per hour and a coefficient of determination of 0.987.

The spread between zones requires explanation rather than omission. HOD Rooms report 25.8 per cent but average only 0.29 kWh per hour, so an absolute error of 0.10 kWh per hour is operationally negligible and reads as a large percentage only because the denominator is small; the zone is 5.7 per cent of consumption. Classroom error concentrates at 07:00 and 17:00, the transition hours when fan and lighting use is governed by occupant behaviour rather than schedule or temperature.

TABLE VII. HOUR-LEVEL DETECTION AGAINST GROUND TRUTH AT INCREASING THRESHOLDS, AND EVENT-LEVEL RECALL.

k	Precision	Recall	F1	Events
3	0.063	0.415	0.110	9 / 11
4	0.096	0.378	0.154	9 / 11
5	0.143	0.336	0.201	9 / 11
6	0.268	0.305	0.285	9 / 11
8	0.532	0.237	0.328	9 / 11
10	0.825	0.177	0.291	9 / 11

Nine of the eleven events were surfaced at every threshold examined; the two missed are an overnight-load event in HOD Rooms and a holiday-consumption event in Classrooms, both in zones whose small absolute

consumption places the injected excess within normal residual spread. Hour-level precision is poor at the operating threshold of three standard deviations, at 0.063, rising to 0.825 at ten standard deviations as recall falls to 0.177. Most false positives arise in Server Room, Classrooms and HOD Rooms, the three zones with the smallest absolute loads.

The reference predictor surfaces nine of the eleven labelled events. At the operating threshold of three standard deviations, hour-level precision of 0.063 implies an operator would examine roughly fifteen false alarms for each true one. These figures characterise this configuration rather than estimating detection performance in general.

D. Cost Structure

TABLE VIII. MODELLED ANNUAL COST COMPOSITION USING THE INDICATIVE CONSTANTS OF TABLE V.

Component	Annual	Share
Energy (147,155 kWh at ₹8.05)	₹11,84,598	49.6%
Maximum demand (173.1 kVA at ₹580)	₹12,04,672	50.4%
Combined	₹23,89,270	100%

Slightly more than half the modelled annual bill is determined by the single highest demand interval. A flat energy-rate model omits the demand-charge component and may therefore produce a different intervention ranking, since measures reducing consumption uniformly act on 49.6 per cent of the modelled bill while measures reducing the peak act on the remaining 50.4 per cent; the two rankings were not computed separately here. Peak coincidence analysis attributes the recorded maximum of 173.1 kVA, against a sanctioned 150 kVA, to both laboratories being in session simultaneously on a warm afternoon. Offsetting one session by an hour requires no expenditure and reduces no consumption, yet lowers the billed peak.

E. Objective Mapping

Table IX maps each objective to the evidence bearing on it and the outcome reached.

TABLE IX. OBJECTIVES MAPPED TO EVIDENCE AND OUTCOME.

Obj.	Evidence	Outcome
O1	Table VI, Fig. 2	Met; 5.6–25.8% MAPE by zone
O2	Section III-F worked example	Partially met; hypothesis set, not diagnosis
O3	Table VIII	Met; demand is 50.4% of modelled cost
O4	Table III, implemented policy	Met; five provenance states
O5	Tables VII and VIII	Met; baseline reproducible, detection weak

V. DISCUSSION

A. Interpretation

The results support a claim that is architectural rather than algorithmic. Forecasting accuracy is not the product; it is the reference that makes a residual meaningful. The residual is useful because the inventory converts it into equipment, the equipment hypothesis because the tariff prices it, and the price because it orders interventions otherwise incommensurable. The demand-charge result illustrates this: a flat energy-rate model omits the demand component entirely and may therefore rank a consumption-reducing measure above a peak-reducing one, although the two rankings were not computed separately in this study.

The detection results qualify this. The pipeline functions as designed and surfaces most injected events, but hour-level precision with the reproducible reference predictor is too low for unattended operation. The trained models achieve lower forecast error than the profile used for this evaluation, but whether that translates into higher detection precision was not tested. Connecting scheduled model inference to the served path and re-running this evaluation is a direct engineering step for future work.

B. Comparison with Prior Work

Against the forecasting literature [1]–[6] this work adds nothing and adopts freely; 5.75 per cent block-level MAPE is unremarkable beside published results on metered data, and no superiority is claimed. The contribution relative to the detection literature [7]–[9] is what follows the flag rather than the flag itself. Relative to the explainability stream [10]–[12] the distinction is ontological: feature attribution explains the model, inventory attribution explains the building, and the two are complementary.

On evidence strength the comparison is unfavourable and should be stated as such. Studies working with metered data [5], [6], [13] evaluate against consumption that actually occurred, with faults that actually happened. This work evaluates against a generated process with injected faults. The accuracy figures are therefore not comparable with those studies in kind, and the detection figures characterise one generator rather than a building. What the synthetic design buys is labelled ground truth, which those studies generally lack; what it costs is external validity.

C. What Is Not Demonstrated

- No verified energy or cost saving. No intervention was implemented and no post-intervention measurement taken. Reported savings are modelled consequences of modelled faults.
- No validation against metered data. Reported accuracy measures the models' ability to learn a physically motivated generative process.
- No confirmed attribution. The procedure narrows causes to a physically admissible set; it cannot establish which is correct.
- No live model inference in the deployed service. The forecast endpoint returns a day-type reference

profile, because the trained models require a 38-feature matrix built offline. Reported model metrics come from that offline evaluation.

- No general detection claim and no statistical significance. Table VII characterises one configuration against eleven events from a single generator, and point estimates are reported without confidence intervals.
- No algorithmic novelty. Every component is standard; the contribution is composition and the interfaces between stages.

D. Threats to Validity and Limitations

Construct validity is the principal threat. Every anomaly label originates in the generator that produced the series, so the detector is scored against a definition of abnormality the data was built to contain; a fault pattern the generator does not model would not appear in the evaluation at all. Internal validity is protected by chronological partitioning and by excluding anomalous hours from training, though the profile feature is fitted on training data and applied to test data, so drift in day-type behaviour across the boundary would be absorbed silently. External validity is limited to one block of one institution; inventory, tariff and calendar are site-specific.

Three further constraints bound the work: tariff constants are indicative published values, so absolute currency figures are provisional while component ratios are robust; detection rests on eleven events; and the server-room inventory is an assumption rather than an audited count. As a deployment rather than scientific limitation, the interface loads styling from a content delivery network.

A note on reproducibility: the baseline in Table VI was recomputed by the method stated in Section III-H, because figures reported in earlier project material could not be reproduced from the dataset.

VI. CONCLUSION AND FUTURE WORK

This paper presented a framework in which energy forecasting serves as the reference for detection, detection output is bound to an audited equipment inventory, and the result is priced over a complete high-tension tariff so interventions can be ranked in currency. On a synthesised year of hourly data for a five-zone academic block, zone-level gradient boosting improved on a recomputed seasonal-naïve baseline in every zone, nine of eleven labelled faults were surfaced by a reproducible reference predictor whose hour-level precision was nonetheless poor, and tariff modelling attributed 50.4 per cent of modelled annual cost to the maximum-demand component.

Two steps follow directly. Circuit-level metering would supply measured data for validation and the granularity to confirm rather than shortlist an attribution; the reconciliation layer already treats a metered feed as a further input path, so

it can be accepted without redesign. A future evaluation should connect scheduled model inference to the served path and measure whether detection precision changes.

REFERENCES

- [1] A. Machado et al., "Statistical and machine learning approaches for electrical energy forecasting," *WIREs Data Mining and Knowledge Discovery*, 2025, doi: 10.1002/widm.70033.
- [2] H. Gao, W. Kang, and M. Fan, "A triple-optimized extreme learning machine model for power load forecasting," *IEEE Open Access Journal of Power and Energy*, vol. 12, 2025, doi: 10.1109/OAJPE.2025.3585994.
- [3] L. Liu, F. H. Juwono, W. K. Wong, and H. Liu, "Building energy consumption prediction: A machine learning approach with feature selection," in *Proc. 10th Int. Conf. Smart Computing and Communication (ICSCC)*, 2024, pp. 159–164, doi: 10.1109/ICSCC62041.2024.10690314.
- [4] Q. Qiao, A. Yunusa-Kaltungo, and R. E. Edwards, "Developing a machine learning based building energy consumption prediction approach using limited data," *Energy Reports*, vol. 9, pp. 3643–3660, 2023, doi: 10.1016/j.egy.2023.02.046.
- [5] M. K. Kim, Y.-S. Kim, and J. Srebric, "Predictions of electricity consumption in a campus building using occupant rates and weather elements with sensitivity analysis," *Sustainable Cities and Society*, vol. 62, Art. no. 102385, 2020, doi: 10.1016/j.scs.2020.102385.
- [6] C. Miller et al., "The Building Data Genome Project 2: Energy meter data from the ASHRAE Great Energy Predictor III competition," *Scientific Data*, vol. 7, no. 1, 2020, doi: 10.1038/s41597-020-00712-x.
- [7] Y. Himeur, K. Ghanem, A. Alsalemi, F. Bensaali, and A. Amira, "Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives," *Applied Energy*, vol. 287, Art. no. 116601, 2021, doi: 10.1016/j.apenergy.2021.116601.
- [8] D. B. Araya, K. Grolinger, H. F. ElYamany, M. A. M. Capretz, and G. Bitsuamlak, "An ensemble learning framework for anomaly detection in building energy consumption," *Energy and Buildings*, vol. 144, pp. 191–206, 2017, doi: 10.1016/j.enbuild.2017.02.058.
- [9] Y. K. Telila, D. Senevirathne, D. Tissera, A. Narayan, M. A. M. Capretz, and K. Grolinger, "Federated learning for anomaly detection in energy consumption data: Assessing the vulnerability to adversarial attacks," *arXiv:2502.05041*, 2025.
- [10] M. Noorchenarboo and K. Grolinger, "Explaining deep learning-based anomaly detection in energy consumption data by focusing on contextually relevant data," *arXiv:2501.06099*, 2025.
- [11] J. Moon, S. Rho, and S. W. Baik, "Toward explainable electrical load forecasting of buildings: A comparative study of tree-based ensemble methods with Shapley values," *Sustainable Energy Technologies and Assessments*, vol. 54, Art. no. 102888, 2022, doi: 10.1016/j.seta.2022.102888.
- [12] W. J. Chung and C. Liu, "Analysis of input parameters for deep learning-based load prediction for office buildings in different climate zones using explainable artificial intelligence," *Energy and Buildings*, vol. 276, Art. no. 112521, 2022, doi: 10.1016/j.enbuild.2022.112521.
- [13] R. Dharani, M. Balasubramonian, T. S. Babu, and B. Nastasi, "Load shifting and peak clipping for reducing energy consumption in an Indian university campus," *Energies*, vol. 14, no. 3, Art. no. 558, 2021, doi: 10.3390/en14030558.
- [14] G. N. Patil and S. S. Tanavade, "Eco-friendly energy efficient classrooms and sustainable campus strategies: A case study on energy management and carbon footprint reduction," *International Journal of Energy Economics and Policy*, vol. 14, no. 3, pp. 188–197, 2024, doi: 10.32479/ijee.15712.
- [15] Z. Li, W. Sun, Y. Xiang, C. Thomson, and G. Harrison, "Arranging university semester date to minimize annual CO2 emission: A UK university case study," *iScience*, vol. 24, no. 12, Art. no. 103414, 2021, doi: 10.1016/j.isci.2021.103414.